

LAKSHYA

JEE 2027

Lecture No.- 24

MATHEMATICS

RELATIONS & FUNCTIONS

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Topics *to be covered*



1 Questions on Composite Function

2 Inverse of a Function

3 Question Practice

4



Question Practice

Ans: 31

31

Critical Thinking Question [JEE Mains-2021]



490 fnt



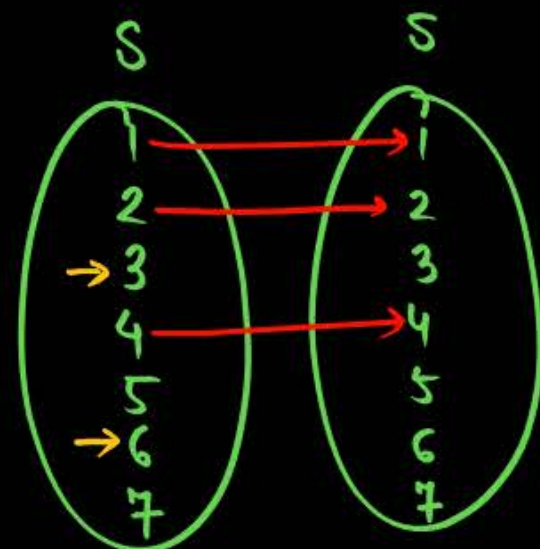
Let $S = \{1, 2, 3, 4, 5, 6, 7\}$. Then the number of possible function $f : S \rightarrow S$ such that $f(m \cdot n) = f(m) \cdot f(n)$ for every $m, n \in S$ and $m \cdot n \in S$ is equal to

$m = n = 1$

$$f(1) = f(1) \cdot f(1) \Rightarrow f(1) = 1 \quad \checkmark$$

$$\begin{array}{l} 1 \times 1 = 1 \\ 2 \times 2 = 4 \\ 2 \times 3 = 6 \end{array}$$

$$1 \times 5 = 5$$



$m = n = 2$

$$f(4) = (f(2))^2$$

$f(2) = 1, f(4) = 1$

$f(2) = 2, f(4) = 4$

$m = 2, n = 3$

$$f(6) = f(2) \cdot f(3)$$

$f(2) = 1$

$f(6) = f(3)$

$f(2) = 2$

$f(6) = 2f(3)$

$$\begin{array}{l} f(6) = 2f(3) \\ = 2 \times 1 \\ = 2 \times 2 \\ = 2 \times 3 \end{array}$$

C- (I) $1 \times 1 \times 1 \times \underbrace{7}_{3 \times 6} \times 7 \times 7 = 7^3 = 343$

C- (II) $1 \times 1 \times 1 \times \underbrace{3}_{f(3)} \times 7 \times 7 = 147$

Question [JEE Mains-2022]



Let $f(x)$ and $g(x)$ be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$, and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is _____.

18
fnf

$$\begin{cases} f(x) = ax^2 + bx + c \\ g(x) = px + q \end{cases}$$

$$g(f(x)) = p(ax^2 + bx + c) + q$$

$$4x^2 + 6x + 1 = pa x^2 + \underline{bp} x + \underline{cp + q}$$

coeff comp.

$$ap = 4, bp = 6, cp + q = 1$$

$$f(g(x)) = a(g(x))^2 + b g(x) + c$$

$$= a(px + q)^2 + b(px + q) + c = 8x^2 - 2x$$

coeff. compare.

$$8 = ap^2, -2 = 2apq + bp, aq^2 + bq + c = 0$$



LAST CLASS REVISION



Answer the following Questions on Composite Functions:

1. If $f(x) = x^3 + 3$ and $g(x) = \sqrt{x}$ then:

(a) $f \circ g = f(g(x)) = x^{3/2} + 3 = (f \circ g)(x)$

(b) $g \circ f = g(f(x)) = \sqrt{x^3 + 3} = (g \circ f)(x)$

2. Range of $g(f(x)) \subseteq$ range of $g(x)$. (T)

3. Domain of $g(f(x)) \subseteq$ domain of $f(x)$. (T)

ex: $\sin(\ln x)$



LAST CLASS REVISION



Notes **

Answer the following Questions on Composite Functions:

4. If f & g are two bijective functions then $f \circ g$ and $g \circ f$ are also bijective. → (T)

5. If $f(g(x))$ is bijective, ^{then} $g(x)$ should be one-one & $f(x)$ should be onto. (T)

↓ inside ↓ outside

Answer the forewing questions:

(ii) If $f(x) = \frac{1}{x}$, $h(x) = \frac{1}{4x^2-1}$ & $g(x) = \frac{5x}{x+2}$, then domain of $g \circ h \circ f(x)$ is ____.

$$f(x) \neq \pm \frac{1}{2}$$

$$h(f(x)) = \frac{1}{4 \times \frac{1}{x^2} - 1}, \quad x \neq 0, \quad x \neq \pm 2$$

$$x \in \mathbb{R} - \{0, \pm 2, \pm 2\sqrt{2}\}$$

$$g(h(f(x))) = \frac{5 h(f(x))}{h(f(x)) + 2}, \quad h(f(x)) \neq -2$$

$$\frac{x^2}{4-x^2} \neq -2$$

$$x^2 \neq -8 + 2x^2$$

$$8 \neq x^2$$

$$x \neq \pm 2\sqrt{2}$$



COMPOSITE OF NON-UNIFORMLY DEFINED FUNCTIONS



To define $f(g(x))$:

- s-(I) Put " $g(x)$ " in $f(x)$ wherever ' x ' is present.**
- s-(II) Draw graph of inside function (i.e. $g(x)$).**
- s-(III) "See graph of $g(x)$ ", remove $g(x)$ by expression & interval of ' x '.**

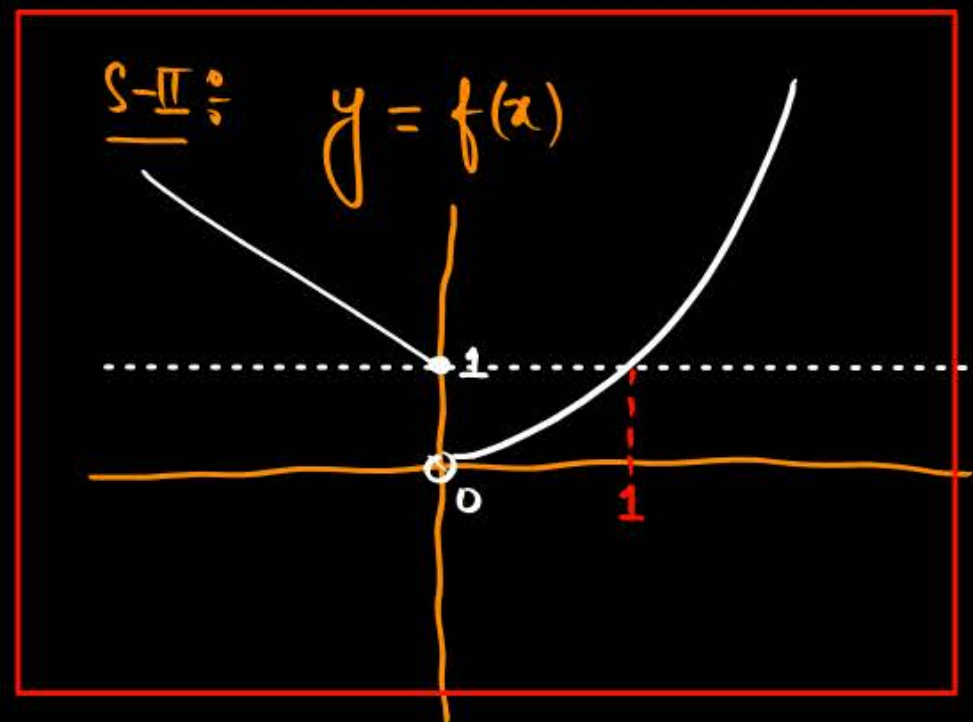
Question



Given $f(x) = \begin{cases} 1-x, & \text{if } x \leq 0 \\ x^2, & \text{if } x > 0 \end{cases}$ and $g(x) = \begin{cases} -x, & \text{if } x < 1 \\ 1-x, & \text{if } x \geq 1 \end{cases}$, find $f \circ g$ and $g \circ f$.

S-I:
$$g(f(x)) = -f(x), \quad \underline{f(x) < 1}$$
$$= 1 - f(x), \quad \underline{f(x) \geq 1}$$

S-II:
$$g(f(x)) = -f(x), \quad x \in (0, 1)$$
$$= 1 - f(x), \quad x \in (-\infty, 0] \cup [1, \infty)$$



$$\Rightarrow f(x) < 1 \Rightarrow x \in (0, 1)$$
$$\Rightarrow f(x) \geq 1 \Rightarrow x \in (-\infty, 0] \cup [1, \infty)$$

$g(f(x))$

S-III

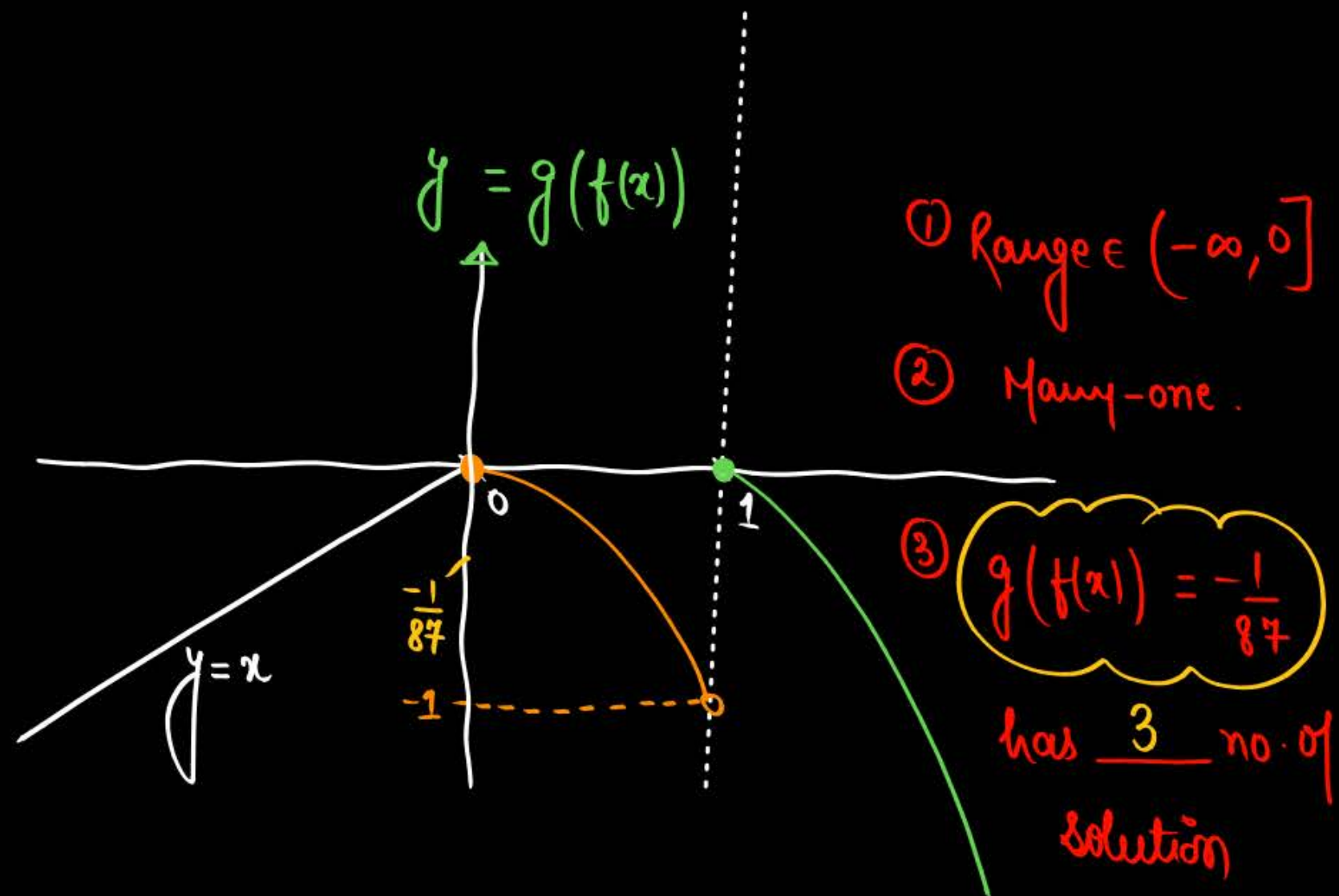
$$g(f(x)) = -x^2, \quad x \in (0, 1)$$

$$= 1 - (1-x), \quad x \in (-\infty, 0]$$

$$= 1 - x^2, \quad x \in [1, \infty)$$

OR

$$\begin{aligned} g \circ f(x) &= -x^2, & x \in (0, 1) \\ &= x, & x \in (-\infty, 0] \\ &= 1 - x^2, & x \in [1, \infty) \end{aligned}$$



Question



Given $f(x) = \begin{cases} 1-x, & \text{if } x \leq 0 \\ x^2, & \text{if } x > 0 \end{cases}$ and $g(x) = \begin{cases} -x, & \text{if } x < 1 \\ 1-x, & \text{if } x \geq 1 \end{cases}$, find $f \circ g$ and $g \circ f$.

M-2

S-I:

$$g(f(x)) = -f(x) \quad , \quad \boxed{f(x) < 1} \\ = 1 - f(x) \quad , \quad \underline{f(x) \geq 1}$$

$$x \in (0, 1)$$

$$x \in (-\infty, 0] \cup [1, \infty)$$

$$f(x) \geq 1$$

$$\begin{cases} \boxed{x \leq 0} \\ 1-x \geq 1 \\ 0 \geq x \end{cases}$$

$$x \in (-\infty, 0]$$

$$\boxed{x > 0} \quad \checkmark$$

$$x^2 \geq 1$$

$$x \in (-\infty, -1] \cup [1, \infty) \quad \checkmark$$

$$x \in [1, \infty)$$

Question [JEE Mains-2024]



If $f(x) = \begin{cases} 2 + 2x, & -1 \leq x < 0 \\ 1 - \frac{x}{3}, & 0 \leq x \leq 3 \end{cases}$; $g(x) = \begin{cases} -x, & -3 \leq x \leq 0 \\ x, & 0 < x \leq 1 \end{cases}$, then range of $(f \circ g(x))$ is

A $(0, 1]$

$f \circ g(x) = 2 + 2g(x)$, $-1 \leq g(x) < 0$ $\rightarrow \phi$

$= 1 - \frac{g(x)}{3}$, $0 \leq g(x) \leq 3$

B $[0, 3)$

C $[0, 1]$

D $[0, 1)$

$-1 \leq g(x) < 0$

$-3 \leq x \leq 0$ $\rightarrow -1 \leq -x < 0$
 $1 \geq x > 0$
OR
 $0 < x \leq 1$
 ϕ

$0 < x \leq 1$ $\rightarrow -1 \leq x < 0$
 ϕ

Question [JEE Mains-2024]



If $f(x) = \begin{cases} 2 + 2x, & -1 \leq x < 0 \\ 1 - \frac{x}{3}, & 0 \leq x \leq 3 \end{cases}$; $g(x) = \begin{cases} -x, & -3 \leq x \leq 0 \\ x, & 0 < x \leq 1 \end{cases}$, then range of $(f \circ g(x))$ is

A $(0, 1]$

B $[0, 3)$

C $[0, 1]$

D $[0, 1)$

$$\begin{aligned} f \circ g(x) &= 2 + 2g(x), \quad -1 \leq g(x) < 0 \\ &= 1 - \frac{g(x)}{3}, \quad 0 \leq g(x) \leq 3 \end{aligned}$$

ϕ

$$x \in [-3, 0] \cup (0, 1]$$

$$\begin{aligned} &0 \leq g(x) \leq 3 \\ &\begin{cases} -3 \leq x \leq 0 \\ 0 < x \leq 1 \end{cases} \\ &\begin{cases} 0 \leq -x \leq 3 \\ 0 \leq x \leq 3 \end{cases} \\ &\begin{cases} 0 \geq x \geq -3 \\ 0 < x \leq 1 \end{cases} \\ &\downarrow \\ &x \in [-3, 0] \end{aligned}$$

Question [JEE Mains-2024]



If $f(x) = \begin{cases} 2 + 2x, & -1 \leq x < 0 \\ 1 - \frac{x}{3}, & 0 \leq x \leq 3 \end{cases}$; $g(x) = \begin{cases} -x, & -3 \leq x \leq 0 \\ x, & 0 < x \leq 1 \end{cases}$, then range of $(f \circ g(x))$ is

A $(0, 1]$

B $[0, 3)$

C $[0, 1]$

D $[0, 1)$

$$\begin{array}{ll} x \in [-3, 0] & x \in (0, 1] \\ \frac{x}{3} \in [-1, 0] & -\frac{1}{3}x \in [-\frac{1}{3}, 0) \\ \frac{x}{3} + 1 \in [0, 1] & 1 - \frac{x}{3} \in [\frac{2}{3}, 1) \end{array}$$

$$\begin{aligned} f(g(x)) &= \left(1 + \frac{x}{3}\right), \quad x \in [-3, 0] \\ &= \left(1 - \frac{x}{3}\right), \quad x \in (0, 1] \end{aligned}$$

Arrows indicate the mapping of the ranges $[0, 1]$ and $[\frac{2}{3}, 1)$ to the corresponding parts of the piecewise function.

Question [JEE Mains-2024]



If $f(x) = \begin{cases} 2 + 2x, & -1 \leq x < 0 \\ 1 - \frac{x}{3}, & 0 \leq x \leq 3 \end{cases}$; $g(x) = \begin{cases} -x, & -3 \leq x \leq 0 \\ x, & 0 < x \leq 1 \end{cases}$, then range of $(f \circ g(x))$ is

A $(0, 1]$

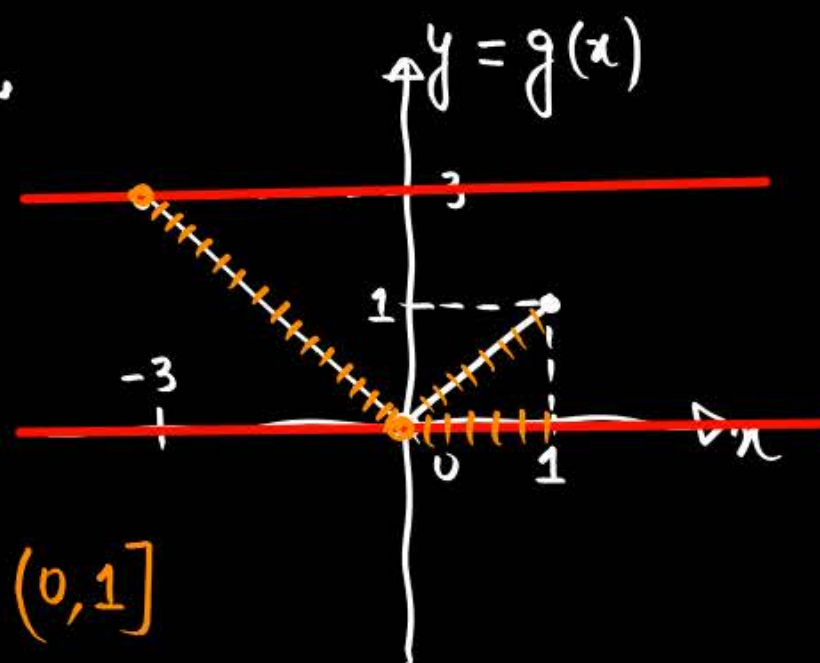
B $[0, 3)$

☒ C $[0, 1]$

D $[0, 1)$

$f \circ g = 2 + 2g(x), \quad -1 \leq g(x) < 0$
 $= 1 - \frac{g(x)}{3}, \quad 0 \leq g(x) \leq 3$

$x \in [-3, 0] \cup (0, 1]$



Question [JEE Mains-2024]



Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as: $f(x) = |x - 1|$ and $g(x) = \begin{cases} e^x, & x \geq 0 \\ x + 1, & x \leq 0 \end{cases}$.

Then the function $f(g(x))$ is

☒ **A** neither one-one nor onto.

☐ **B** one-one but not onto.

☐ **C** both one-one and onto.

☐ **D** onto but not one-one.

$$f(g(x)) = |g(x) - 1|$$

$$\# f \circ g(x) = |e^x - 1|, \quad x \geq 0$$

$$= |\underline{x+1} - 1|, \quad x \leq 0.$$

$$\# \mathbb{R} \rightarrow (\mathbb{R})$$

into

$$f \circ g(x) = |e^x - 1|, \quad x \geq 0$$

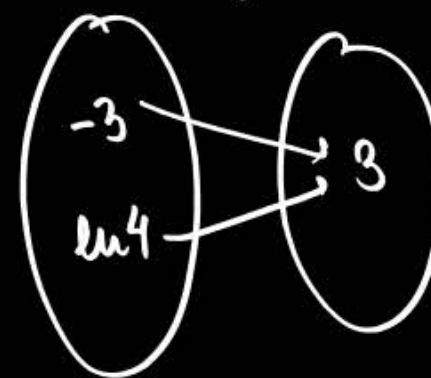
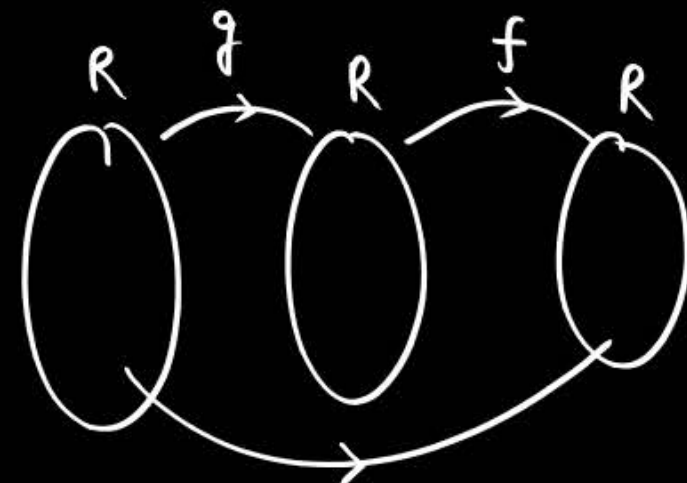
$$= |x|, \quad x \leq 0$$

$$|e^x - 1| = 3 \quad |x| = 3$$

$$e^x - 1 = 3 \quad \downarrow$$

$$e^x = 4 \quad x = -3$$

$$\hookrightarrow x = \ln 4.$$

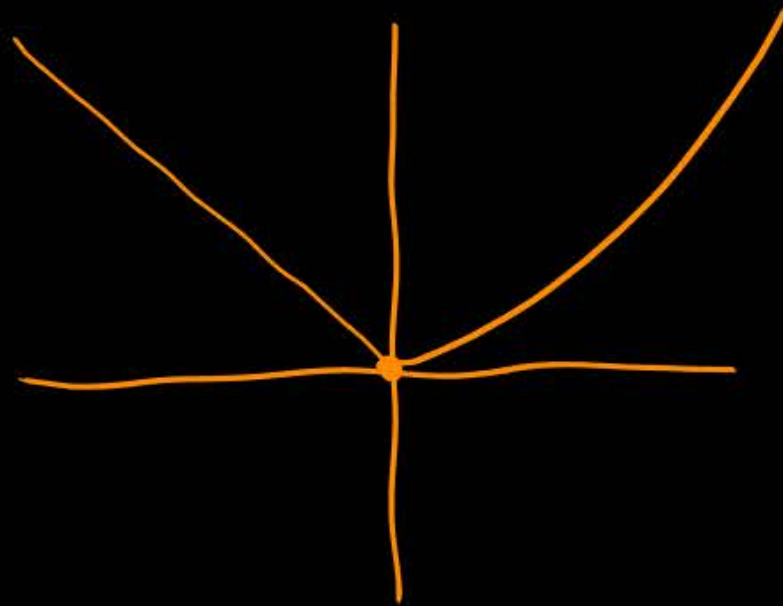
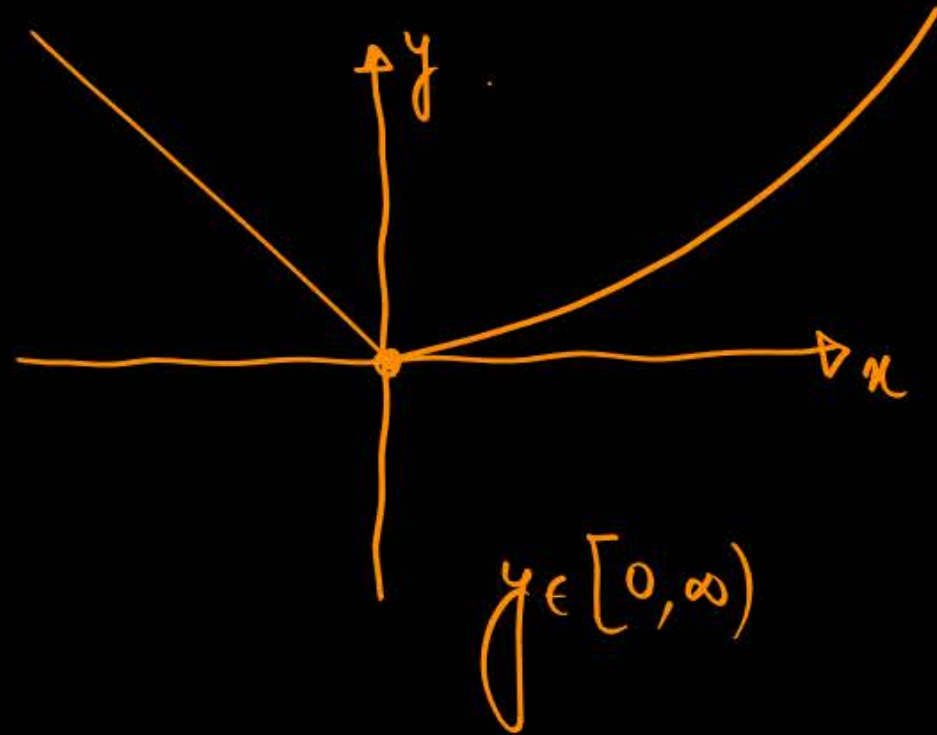


$$\# \text{fog}(x) = |e^x - 1|, x \geq 0$$

$$= |x|, x \leq 0$$

$$\text{fog} = e^x - 1, x \geq 0$$

$$= -x, x \leq 0$$



$$\boxed{\frac{dy}{dx}} =$$

$$e^x$$

+ve ✓

$$= -1$$

-ve ✓

Many-one.

Question [JEE Mains-2022]

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Define $f: S \rightarrow S$ as

$$f(n) = \begin{cases} 2n, & \text{if } n = 1, 2, 3, 4, 5 \\ 2n - 11, & \text{if } n = 6, 7, 8, 9, 10 \end{cases}$$

Let $g: S \rightarrow S$ be a function such that $f \circ g(n) = \begin{cases} n + 1, & \text{if } n \text{ is odd} \\ n - 1, & \text{if } n \text{ is even} \end{cases}$,

then $g(10)((g(1) + g(2) + g(3) + g(4) + g(5)))$ is equal to

10

Question [JEE Mains-2021]



Let $f, g: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(n+1) = f(n) + f(1) \forall n \in \mathbb{N}$ and g be any arbitrary function. Which of the following statements is NOT true?

- A** If $f \circ g$ is one - one, then g is one - one
- B** f is one-one
- C** If g is onto, then $f \circ g$ is one-one
- D** If f is onto, then $f(n) = n \forall n \in \mathbb{N}$

Ans.

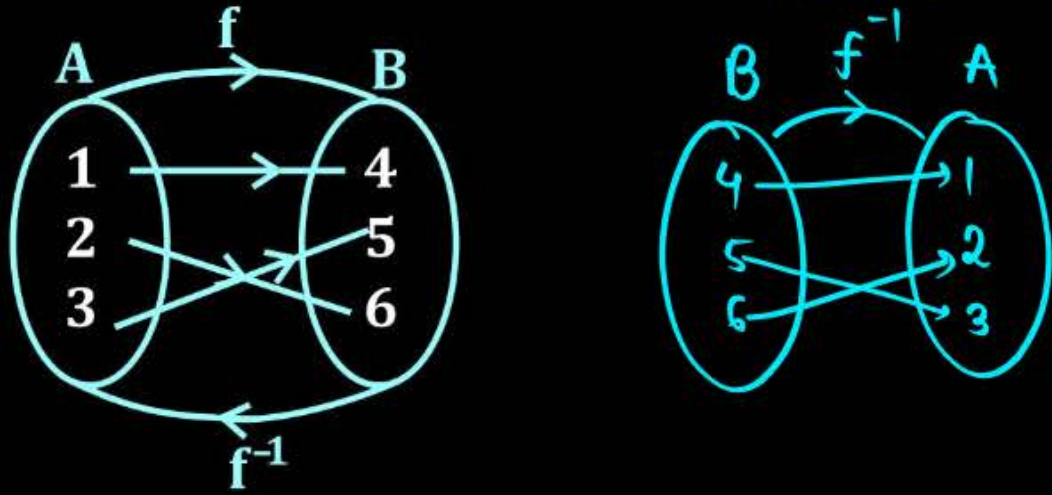


INVERSE OF FUNCTION



Represent: $f^{-1}(x)$ $f^{-1}(x) \neq \frac{1}{f(x)}$

Note:

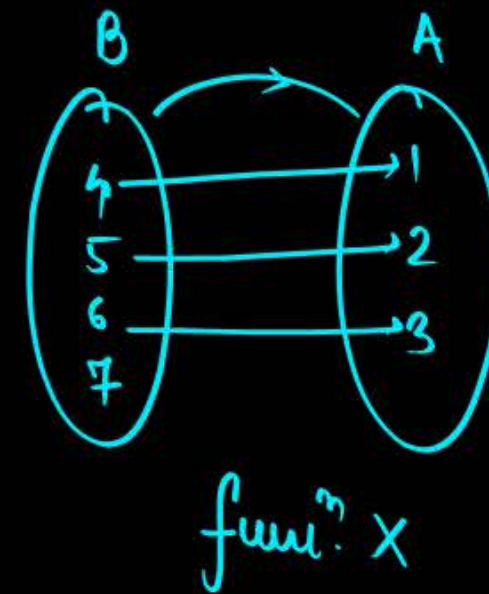
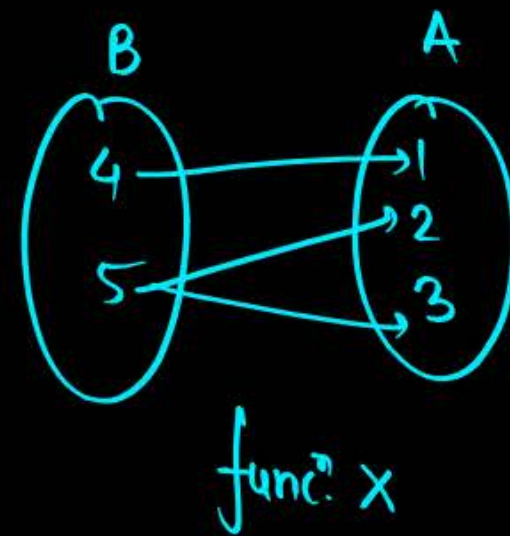
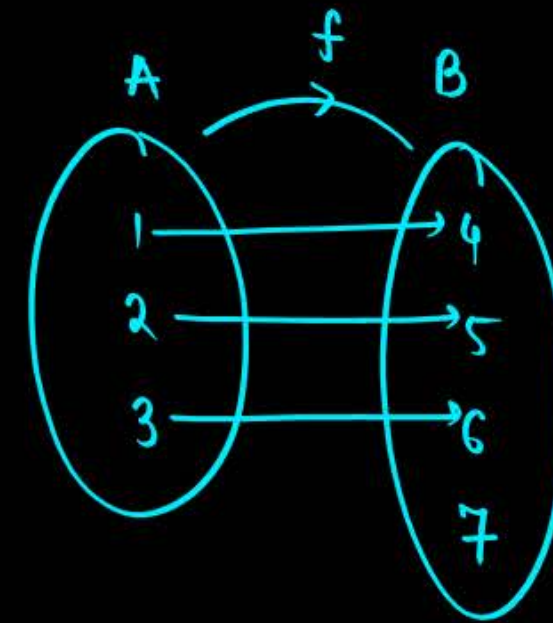
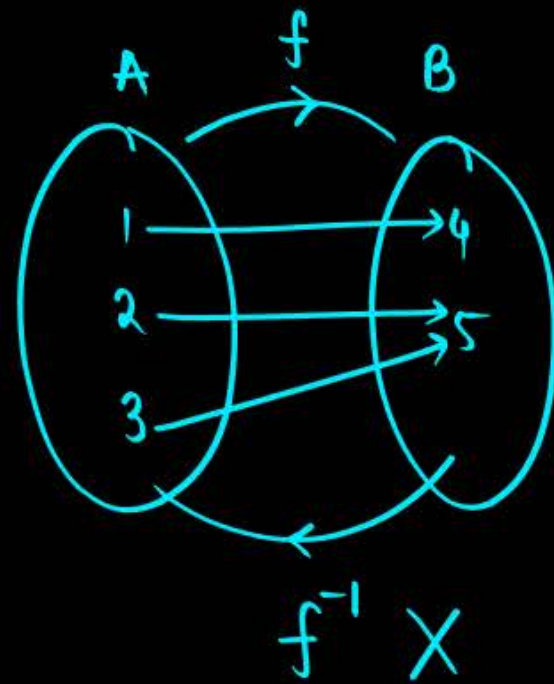


Domain of $f = \{1, 2, 3\} = \text{Range of } f^{-1}$

Range of $f = \{4, 5, 6\} = \text{Domain of } f^{-1}$

$f = \{(1, 4), (2, 6), (3, 5)\}$

$f^{-1} = \{(4, 1), (6, 2), (5, 3)\}$





HOW TO FIND INVERSE??



S-I: Check given function is **BIJECTIVE** or not.

S-II: Write $y = f(x)$ & **interchange x & y** .

S-III: Now obtain ' y ' alone in terms of ' x '.

S-IV: In case of multiple answer, use point verification method.

ex:

$$f: \overbrace{(0, \infty)}^D \rightarrow \overbrace{\mathbb{R}}^{\text{Range.}}$$

$$f(x) = \ln x \quad \checkmark$$

$f^{-1}(x) = ??$

Solⁿ:

$$y = \ln x$$

$$x \leftrightarrow y$$

$$x = \ln y$$

$$e^x = y$$

$$f^{-1}(x) = e^x$$

$$\mathbb{R} \rightarrow (0, \infty)$$

RAPID FIRE ROUND



Find inverse of following functions :

(i) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \ln(x + \sqrt{1+x^2})$ (Bijective)

(ii) $f: [1, \infty) \rightarrow [2, \infty), f(x) = x + \frac{1}{x}$ [JEE-2001, 1M]

(iii) $f: [1, \infty) \rightarrow [1, \infty), f(x) = 2^{x(x-1)}$ [JEE-1999, 2M]

sol.

(i) $y = \ln(x + \sqrt{1+x^2})$

$x \leftrightarrow y$
 $x = \ln(y + \sqrt{1+y^2})$

$e^x = y + \sqrt{y^2+1}$

$(e^x - y)^2 = y^2 + 1$

$e^{2x} - 2ye^x + y^2 = y^2 + 1$

$e^{2x} - 1 = 2ye^x$

$\frac{1}{2} \left(e^x - \frac{1}{e^x} \right) = y$

$\mathbb{R} \rightarrow \mathbb{R}$

or $\sqrt{y^2+1} + y = e^x$

$\sqrt{y^2+1} - y = \frac{1}{e^x}$

$\ominus$
 $2y = e^x - \frac{1}{e^x}$

②

$$f: \overbrace{[1, \infty)}^D \rightarrow \overbrace{[2, \infty)}^R, \quad f(x) = x + \frac{1}{x}$$

$$f(x) = x + \frac{1}{x}$$

$$y = x + \frac{1}{x}$$

$$x \leftrightarrow y$$

$$x = y + \frac{1}{y}$$

$$xy = y^2 + 1$$

$$y^2 - xy + 1 = 0$$

$$y = \frac{x \pm \sqrt{x^2 - 4}}{2}$$

$$\# y = \frac{x + \sqrt{x^2 - 4}}{2}$$

$$\times y = \frac{x - \sqrt{x^2 - 4}}{2}$$

$$x = \frac{5}{2} \rightarrow y = \frac{\frac{5}{2} - \frac{3}{2}}{2} = \frac{1}{2}$$

$$\left(\frac{5}{2}, \frac{1}{2}\right)$$

$$x = \frac{5}{2} \rightarrow y = \frac{\frac{5}{2} + \sqrt{\frac{25}{4} - 4}}{2} = \frac{4}{2} = 2$$

$$\downarrow \left(\frac{5}{2}, 2\right)$$

$$\# f^{-1}: [2, \infty) \rightarrow [1, \infty), \quad f^{-1}(x) = \frac{x + \sqrt{x^2 - 4}}{2}$$

Question [JEE Mains-2020]



The inverse function of $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$, $x \in (-1, 1)$ is

A $\frac{1}{4} \log_e \left(\frac{1+x}{1-x} \right)$

B $\frac{1}{4} \log_e \left(\frac{1-x}{1+x} \right)$

C $\frac{\log_8 e}{4} \log_e \left(\frac{1+x}{1-x} \right)$

D $\frac{\log_8 e}{4} \log_e \left(\frac{1-x}{1+x} \right)$

Ans.

DO IT BY YOURSELF (DIBY)

NUMBER OF FUNCTION CASE MAKING / CRITICAL THINKING

DIBY-160

Let $R = \{a, b, c, d, e\}$ and $S = \{1, 2, 3, 4\}$. Total number of onto function $f: R \rightarrow S$ such that $f(a) \neq 1$, is equal to
[JEE Mains-2023]
[Ans. : 180]

DIBY-161

Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16\}$. Then the number of many-one functions $f: A \rightarrow B$ such that $1 \in f(A)$ is equal to
[JEE Mains-2025]
 (A) 163 (B) 139 (C) 151 (D) 127
[Ans. : (c)]

DIBY-162

Consider two sets $A = \{x \in \mathbb{Z}: |(|x - 3| - 3)| \leq 1\}$ and
 $B = \left\{x \in \mathbb{R} - \{1, 2\}: \frac{(x-2)(x-4)}{x-1} \log_e(|x - 2|) = 0\right\}$. Then the number of onto function $f: A \rightarrow B$ is equal to
[JEE Mains-2026]
 (A) 62 (B) 32 (C) 81 (D) 79
[Ans. : (A)]

**NUMBER OF FUNCTION
CASE MAKING / CRITICAL THINKING**



DIBY-163

Let $A = \{(x, y) : 2x + 3y = 23, x, y \in N\}$ and $B = \{x : (x, y) \in A\}$. Then the number of one-one functions from A to B is equal to _____ **[JEE Mains-2024]**

[Ans. : 24]

DIBY-164

The number of functions f , from the set $A = \{x \in N : x^2 - 10x + 9 \leq 0\}$ to the set $B = \{n^2 : n \in N\}$ such that $f(x) \leq (x - 3)^2 + 1$, for every $x \in A$, is ____ **[JEE Mains-2022]**

[Ans. : 1440]

DIBY-165

The number of functions $f: \{1, 2, 3, 4\} \rightarrow \{a, b, c\}$, which are not onto, is:

[JEE Mains-2026]

(A) 48

(B) 45

(C) 51

(D) 35

[Ans. : (B)]

**NUMBER OF FUNCTION
CASE MAKING / CRITICAL THINKING**

DIBY-166

Let $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Then the number of bijective functions $f: A \rightarrow A$ such that $f(1) + f(2) = 3 - f(3)$ is equal to ____

[JEE Mains-2021]

[Ans. : 720]

COMPOSITE FUNCTION



DIBY-167

If $g(x) = 3x^2 + 2x - 3$, $f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to: [JEE Mains-2026]

- (A) $\frac{25}{6}$ (B) $-\frac{7}{2}$ (C) $\frac{7}{2}$ (D) $-\frac{25}{6}$

[Ans.: (C)]

DIBY-168 (Pattern Based)

Let $f(x) = \frac{x-1}{x+1}$, $x \in R - \{0, -1, 1\}$. If $f^{n+1}(x) = f(f^n(x))$ for all $n \in N$, then $f^6(6) + f^7(7)$ is equal to [JEE Mains-2022]

- (A) $\frac{7}{6}$ (B) $-\frac{3}{2}$ (C) $\frac{7}{12}$ (D) $-\frac{11}{12}$

[Ans.: (B)]

DIBY-169 ★★★★★

If $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$ and $(f \circ f)(x) = g(x)$, where $f: R - \left\{\frac{2}{3}\right\} \rightarrow R - \left\{\frac{2}{3}\right\}$, then $(g \circ g \circ g)(4)$ is equal to [JEE Mains-2024]

- (A) $-\frac{19}{20}$ (B) $\frac{19}{20}$ (C) -4 (D) 4

[Ans.: (D)]

COMPOSITE FUNCTION



DIBY-170

If $g(x) = x^2 + x - 1$ and $(g \circ f)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to

[JEE Mains-2020]

(A) $\frac{1}{2}$

(B) $\frac{-3}{2}$

(C) $\frac{-1}{2}$

(D) $\frac{3}{2}$

[Ans.: (C)]

DIBY-171

Let $f(x) = x^2 + 9$, $g(x) = \frac{x}{x-9}$ and $a = f \circ g(10)$, $b = g \circ f(3)$. If e and 1 denote the

eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$, then

$8e^2 + 1^2$ is equal to.

[JEE Mains-2024]

(A) 16

(B) 8

(C) 6

(D) 12

[Ans.: (B)]

DIBY-172

Let $f: R - \left\{\frac{\alpha}{6}\right\} \rightarrow R$ be defined by $f(x) = \frac{5x+3}{6x-\alpha}$. Then the value of α for which $(f \circ f)(x) =$

x , for all $x \in R - \left\{\frac{\alpha}{6}\right\}$, is :

[JEE Mains-2021]

(A) 6

(B) 8

(C) No such α exists

(D) 5

[Ans.: (D)]

DIBY-173

For a suitably chosen real constant a , let a function, $f: R - \{-a\} \rightarrow R$ be defined by $f(x) = \frac{a-x}{a+x}$. Further suppose that for any real number $x \neq -a$ and

$f(x) \neq -a$, $(f \circ f)(x) = x$. Then $f\left(-\frac{1}{2}\right)$ is equal to [JEE Mains-2020]

- (A) -3 (B) $\frac{1}{3}$ (C) $-\frac{1}{3}$ (D) 3

[Ans.: (D)]

DIBY-174

For $x \in R - [0, 1]$, let $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$ and $f_3(x) = \frac{1}{1-x}$ be three given functions. If a function, $J(x)$ satisfies $(f_2 \circ J \circ f_1)(x) = f_3(x)$, then $J(x)$ is equal to

[JEE Mains-2019]

- (A) $f_3(x)$ (B) $\frac{1}{x}f_3(x)$ (C) $f_2(x)$ (D) $f_1(x)$

[Ans.: (A)]

DIBY-175

Let $f(x) = x^2, x \in R$. For any $A \subseteq R$, define $g(A) = \{x \in R: f(x) \in A\}$. If $S = [0, 4]$, then which one of the following statements is not true? [JEE Mains-2019]

- (A) $f(g(S)) \neq f(S)$ (B) $f(g(S)) = S$
 (C) $g(f(S)) = g(S)$ (D) $g(f(S)) \neq S$

[Ans. : (C)]

DIBY-176 (A.O.D.)

Let $f(x) = e^x - x$ and $g(x) = x^2 - x, \forall x \in R$. Then the set of all $x \in R$, where the function $h(x) = (f \circ g)(x)$ is increasing, is: [JEE Mains-2019]

- (A) $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$ (B) $\left[0, \frac{1}{2}\right] \cup [1, \infty)$
 (C) $\left[\frac{-1}{2}, 0\right] \cup [1, \infty)$ (D) $[0, \infty)$

$$\# \frac{dy}{dx} \geq 0$$

[Ans. : (B)]

DIBY-177

For $x \in \left(0, \frac{3}{2}\right)$, let $f(x) = \sqrt{x}$, $g(x) = \tan x$ and $h(x) = \frac{1-x^2}{1+x^2}$. If $\phi(x) = (h \circ f \circ g)(x)$, then

$\phi\left(\frac{\pi}{3}\right)$ is equal to:

[JEE Mains-2019]

- (A) $\tan \frac{11\pi}{12}$ (B) $\tan \frac{\pi}{12}$ (C) $\tan \frac{5\pi}{12}$ (D) $\tan \frac{7\pi}{12}$

[Ans. : (A)]

DIBY-178

Let $f, g: (1, \infty) \rightarrow \mathbb{R}$ be defined as $f(x) = \frac{2x+3}{5x+2}$ and $g(x) = \frac{2-3x}{1-x}$. If the range of the function $\text{fog} : [2, 4] \rightarrow \mathbb{R}$ is $[\alpha, \beta]$, then $\frac{1}{\beta-\alpha}$ is equal to

- (A) 68 (B) 29 (C) 2 (D) 56

[Ans. : (D)]

COMPOSITE FUNCTION



DIBY-179 (Pattern Based)

Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x}{\sqrt{1+9x^2}}$ If the composition of

$f(f \circ f \circ f \circ \dots \circ f)(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the value of $\sqrt{3\alpha + 1}$ is equal to
10 times

[JEE Mains-2024]

[Ans. : 1024]

DIBY-180 (Hold!)

Let α, β and γ be three positive real numbers. Let $f(x) = \alpha x^5 + \beta x^3 + \gamma x, x \in R$ and $g: R \rightarrow R$ be such that $g(f(x)) = x$ for all $x \in R$. If $a_1, a_2, a_3, \dots, a_n$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\frac{1}{n}\sum_{i=1}^n f(a_i)\right)\right)$ is equal to:

[JEE Mains-2022]

(A) 0

(B) 3

(C) 9

(D) 27

[Ans. : (A)]

COMPOSITE FUNCTION



DIBY-181

Let $f: R - \left\{-\frac{1}{2}\right\} \rightarrow R$ and $g: R - \left\{-\frac{5}{2}\right\} \rightarrow R$ be defined as $f(x) = \frac{2x+3}{2x+1}$ and $g(x) = \frac{|x|+1}{2x+5}$.

Then the domain of the function $f \circ g$ is

[JEE Mains-2024]

- (A) $R - \left\{-\frac{5}{2}\right\}$ (B) R (C) $R - \left\{-\frac{7}{4}\right\}$ (D) $R - \left\{-\frac{5}{2}, -\frac{7}{4}\right\}$

[Ans.: (A)]

DIBY-182

Let $f(x) = \log_e x$ and $g(x) = \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$. Then, the domain of $f \circ g$ is

[JEE Mains-2025]

- (A) $(0, \infty)$ (B) $[0, \infty)$ (C) $[1, \infty)$ (D) R

[Ans.: (D)]

DIBY-183

Let $f: R \rightarrow R$ be defined as $f(x) = x - 1$ and $g: R - \{1, -1\} \rightarrow R$ be defined as

$g(x) = \frac{x^2}{x^2 - 1}$. Then the function $f \circ g$ is

[JEE Mains-2022]

- | | |
|------------------------------------|---------------------------------------|
| (A) One-one but not onto function | (B) Onto but not one-one function |
| (C) Both one-one and onto function | (D) Neither one-one nor onto function |

[Ans. : (D)]

DIBY-184

Let $f(x) = \begin{cases} x+1; & x \leq 1 \\ 2x+1; & 1 < x \leq 2 \end{cases}$ and $g(x) = \begin{cases} x^2; & -1 \leq x < 2 \\ x+2; & 2 \leq x \leq 3 \end{cases}$. Find $f \circ g$ and $g \circ f$.

$$[\text{Ans. : } f(g(x)) = \begin{cases} x^2 + 1; & -1 \leq x \leq 1 \\ 2x^2 + 1; & 1 < x \leq \sqrt{2} \end{cases}$$

$$g \circ f(x) = (x+1)^2, -2 \leq x < 1]$$

DIBY-185

Let $f: R \rightarrow R$ and $g: R \rightarrow R$ be defined as $f(x) = \begin{cases} \log_e x, & x > 0 \\ e^{-x}, & x \leq 0 \end{cases}$ and

$g(x) = \begin{cases} x, & x \geq 0 \\ e^x, & x < 0 \end{cases}$. Then $g \circ f: R \rightarrow R$ is

[JEE Mains-2024]

- (A) one-one but not onto
(C) onto but not one-one

- (B) neither one-one nor onto
(D) both one-one and onto

[Ans. : (B)]

DIBY-186

Let $A = \{1, 2, 3, \dots, 10\}$ and $f: A \rightarrow A$ be defined as $f(k) = \begin{cases} k+1, & \text{if } k \text{ is odd} \\ k, & \text{if } k \text{ is even} \end{cases}$. Then the number of possible functions $g: A \rightarrow A$ such that $g \circ f = f$ is **[JEE Mains-2021]**

(A) 10^5 (B) ${}^{10}C_5$ (C) 5^5 (D) $5!$

[Ans. : (A)]



Homework



Re-attempt all good Questions of Class

DIBY

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians